

Order Flow & Pre-Move Detection System

Expert Consultant Review

NSE Equities | Intraday $\geq 5\%$ Move Detection | Microstructure ML

Generated: June 2026 | Data: 20-Level DOM + Tick-Level Feed | Exchange: NSE India

1. Executive Summary & Overall Verdict

The pipeline engineering is genuinely impressive for a first-time microstructure ML project. The contamination catch, formal review/rebuttal loops, shuffle testing, and stock-level CV all reflect correct instincts. However, the project is currently doing ML on a single-day cross-section — that is stock characterization, not alpha research. The binding constraint is data depth, not model complexity.

The restated goal — identifying stocks that will move $\geq 5\%$ from entry to end-of-day — is a far better problem than fixed-horizon classification. It has a clean economic interpretation, maps directly to a tradeable strategy, and aligns with how institutional intraday desks actually think about stock selection.

Dimension	Status	Comment
Pipeline integrity	PASS	Contamination fixed, validation gates at each stage, clean rebuild
Data depth	CRITICAL	Single day = stock characterization, not temporal signal. Need 60+ days.
Labelling approach	REVISE	Fixed-horizon discrete labels lose economic meaning. Switch to MFE-to-close.
Validation realism	REVISE	Stock-level CV is correct for single day but walk-forward is required next.
Trade inference	UNVALIDATED	Two strongest features built on inferred trades never checked against real ticks.
Feature set	PARTIAL	Good foundation. Missing micro-price, book pressure, cumulative delta, OAR.

2. Critical Bugs & Red Flags

2.1 $MCC=0.31$ is likely stock characterization, not alpha

Severity: IMMEDIATE | Action: Test before any further modelling

Top RF features are spread (importance 0.196) and volatility_5m (0.165). These are near-static stock properties — a large-cap blue chip has low spread always; a small-cap always has high spread. The model is almost certainly learning 'volatile stocks move more' which is trivially true and completely unexploitable.

Diagnostic test (20 minutes):

- For each stock compute mean(spread) and mean(volatility_5m) across all its bars
- Use these per-stock means as the only features in a fresh LR model
- If $MCC > 0.15$, your $RF=0.31$ result is a stock classifier — the temporal signal is near zero
- Within-stock temporal CV: split each stock's bars 70/30 by time, train and test on the same stock. If MCC collapses, the signal is cross-sectional.

2.2 Overlapping labels create spurious autocorrelation

Severity: SIGNIFICANT | Action: Switch to MFE labels

At 1-min bars with a 60m horizon, adjacent labels share ~98% of their forward window. This creates massive label autocorrelation. Your stock-level CV prevents temporal leakage across stocks — correct — but with only one day per stock (~374 bars), all bars share the same intraday macro environment. Walk-forward validation (train on morning session, test on afternoon) is the honest single-day test.

2.3 Negative spread values (crossed book) in 6 stocks

Severity: DATA CORRUPTION | Action: Quarantine immediately

A crossed book ($ask1 < bid1$) never exists in real NSE markets. If you're seeing it in cleaned data, your Stage 1 cleaner is inverting bid/ask columns for those instruments. These 6 stocks propagate corrupt values into iceberg_score, imbalance_top5, and all depth features. Quarantine them until the source data is investigated. Check if they are preference shares, ETFs, or SME instruments with non-standard DOM structure.

2.4 Trade inference accuracy unvalidated — contaminates your two best features

Severity: SIGNIFICANT | Action: 2-hour validation task

large_trade_ratio (effect size 0.571) and iceberg_score (0.266) are your strongest features by a wide margin, and both are computed from the inferred trades path — not directly from raw ticks. You have real tick data for all 438 stocks sitting in your pipeline right now. A bar-level comparison takes 2 hours. Target: Pearson $r > 0.85$ on trade count, $r > 0.70$ on directional split. Below these thresholds, your feature set needs a rebuild before acquiring more data.

2.5 vwap_distance of 296.9 is computation error, not outlier

Severity: SIGNIFICANT | Action: Gate on minimum cumulative volume

A vwap_distance of 296.9 means price is 29,690% from VWAP. This is physically impossible intraday. The cause: VWAP starts at 0 or near-zero in the opening minutes before the first trades execute, causing a division by a near-zero denominator. Gate the feature: vwap_distance = NaN for the first N bars until cumulative volume exceeds a minimum threshold (e.g., 500 contracts or 0.1% of average daily volume). This should produce NaN for the opening period, not extreme values that survive RobustScaler.

3. The Right Problem: Intraday $\geq 5\%$ Move Detection

3.1 Why the current labels are wrong for this goal

Your current labels answer: 'Did price exceed 5% at exactly $t+60$ minutes?' The question you actually want to answer is: 'Will price reach +5% at any point between now and market close?' These are very different questions.

Current label:

- $\text{forward_return} = (\text{LTP}[t+60] - \text{LTP}[t]) / \text{LTP}[t]$
- LONG if return $\geq 5\%$ exactly at 60 minutes
- Misses moves that hit 5% at 35 minutes then reversed
- Misses moves that hit 5% at 90 minutes

Target label:

- $\text{MFE_to_close} = \max(\text{high}[t : \text{market_close}] - \text{entry}[t]) / \text{entry}[t]$
- LONG if $\text{MFE_to_close} \geq 0.05$
- SHORT if $\min(\text{low}[t : \text{market_close}] - \text{entry}[t]) / \text{entry}[t] \leq -0.05$
- Economically meaningful: maps directly to your trade parameter
- No information loss from fixed-horizon sampling

3.2 What pre-move stocks have in common — the microstructure signature

From market microstructure research and institutional prop desk practice, stocks that make $\geq 5\%$ intraday moves share a specific signature in the 30-60 minutes before the move begins. The following are ordered by predictive importance based on empirical evidence from liquid equity markets:

Order Flow Signals (Highest Predictive Power)

Signal	What It Looks Like Before a 5% Move
Cumulative delta divergence	Buy volume accumulating while price is flat or barely moving. Someone absorbing the offer without pushing price yet — institutional accumulation before markup. Single strongest pre-move signal.
Large trade ratio acceleration	Ratio going from 0.1 to 0.4 within 15 minutes. The acceleration matters more than the absolute level.
Iceberg detection	Repeated refresh at the same price level after executions. Quiet consumption of the offer before the ask lifts.
Order arrival rate spike	DOM updates per second suddenly doubling or tripling while price is in a tight range. Institutional order placement accelerating.

Book Structure Signals

Signal	What It Looks Like Before a 5% Move
Ask thinning without bid thinning	Offer side quietly disappearing (orders pulled or consumed) while bid stays thick. Book 'setting up' for upside move. Visible across all 20 levels.
Bid stack building	Large orders appearing at progressively higher bid levels. Institutional floor building under the stock.
Book pressure	Imbalance x total depth. 60% bid imbalance with 1,000 lots means something very different from 60% with 10 lots.
Depth slope steepening	Quantity dropping sharply as you move from L1 to L5. Thin book away from best = liquidity trap setup before breakout.

Volume & Price Signals

Signal	What It Looks Like Before a 5% Move
Volume burst with price compression	High volume but price not moving much. Classic Wyckoff accumulation — 'cause' building before 'effect'. Price held in a tight range despite heavy flow.
Trade frequency acceleration	Trades per minute doubling or tripling while price is in a tight range.
Opening range breakout setup	Stock building higher lows within the first 30-minute range. Coiling structure before breakout.

Relative strength vs Nifty	Stock up 2% while Nifty up 0.3% in the first hour. Persistent relative outperformance is a strong leading indicator. Model will confirm importance.
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4. Feature Engineering: Complete List

4.1 Current features — keep or revise

Feature	Action	Notes
large_trade_ratio	KEEP — validate first	ES=0.571 strongest signal. Built on inferred trades — validate against real ticks before trusting.
iceberg_score	KEEP — validate first	ES=0.266. Excellent microstructure signal. Same validation requirement as above.
delta_1m	KEEP	ES=0.488. Core order flow signal. Correct as computed.
imbalance_top5	KEEP	ES=0.121. Best of the three imbalance features.
aggressor_ratio	KEEP	ES=0.111. Rough VPIN proxy — keep until true VPIN is added.
volatility_5m	KEEP — context only	ES=0.057. Likely a stock-type feature. Keep as regime context but do not rely on it as a directional signal.
vwap_distance	FIX first	Gate on minimum cumulative volume. Current values physically impossible (296.9).
spread	INVESTIGATE	RF importance 0.196. Stock-type feature. Check if it adds temporal signal after controlling for per-stock mean spread.
delta_5m	EXCLUDED — correct	r=0.944 with delta_1m. Correct exclusion.
order_cancel_rate	REBUILD	All zeros — current implementation broken. See Section 4.3 for estimation approach.
depth_drop_bid/ask	LOW PRIORITY	ES=0.009/0.006. May recover value when computed as rate-of-change rather than level.

4.2 New features to add — prioritised

Feature	Priority	Formula / Notes
Micro-price	HIGH	$(ask_qty \times bid + bid_qty \times ask) / (bid_qty + ask_qty)$. Better price estimate than mid, accounts for L1 book imbalance. Standard institutional reference price.
Cumulative delta since open	HIGH	Running sum of delta_1m from 9:15am to current bar. Persistent directional pressure. Single most important addition for $\geq 5\%$ move detection.

Book pressure	HIGH	imbalance_top5 x (total_bid_qty + total_ask_qty). Imbalance weighted by book depth.
Time since open (minutes)	HIGH	Minutes elapsed since 9:15am. A 5% move at 9:25am is very different from 1:30pm.
Distance from day high/low	HIGH	(current_price - day_high) / day_high and (current_price - day_low) / day_low. Context for breakout proximity.
Opening range features	HIGH	Opening range = high/low of first 15 minutes. Distance from OR high, OR low, OR midpoint. Breakouts from OR are primary source of >=5% moves.
Cumulative delta divergence	HIGH	delta_1m_10bar_sum vs price_change_10bar. Positive delta with flat/negative price = accumulation signal.
Order arrival rate	MEDIUM	DOM updates per second within the 1-min bar. Acceleration signals institutional order placement. Requires timestamp-level computation before aggregation.
Depth slope	MEDIUM	Gradient of quantity across bid levels L1 to L10. Steep drop = thin book above = breakout setup.
Relative strength vs Nifty	MEDIUM	stock_return_since_open - nifty50_return_since_open. Requires Nifty index feed as additional data source.
VPIN proxy	MEDIUM	Volume-sync probability of informed trading. Bucket last N trades by volume, measure buy/sell imbalance per bucket. More robust than aggressor_ratio.
Ask replenishment absence	MEDIUM	After ask consumed, does it refill quickly? Absence of refill = weak supply = bullish. Complement to iceberg_score.
Trade size distribution	LOW	std(trade_size) and skew(trade_size) within 1-min bar. Institutional trades have different size distribution than retail.
Cancel-to-trade ratio proxy	LOW	qty_drops_without_LTP_change / total_qty_changes. Noisy but better than all-zeros. ~60-70% accuracy on NSE 20-level DOM.

4.3 Estimating order_cancel_rate from DOM

The current implementation produces all zeros because it cannot distinguish cancellations from executions using DOM snapshots alone. A probabilistic approach using DOM-diff logic:

- qty_drop_without_trade: quantity decrease at a level NOT followed by an LTP change within a ~1 tick time window
- new_order_arrival: quantity increase without a corresponding trade at that level
- cancel_rate_proxy = $\text{sum}(\text{qty_drops_without_trade}) / \text{sum}(\text{total_qty_changes})$

Expected accuracy: ~60-70% on NSE. Confounds partial fills with cancellations when trades happen at the same level. Still a meaningful signal vs all-zeros.

5. Labelling Strategy

5.1 Switch to MFE Triple Barrier Labels

The Triple Barrier Method (Lopez de Prado, Advances in Financial ML, Chapter 3) is the industry standard for intraday labels. Instead of fixed-horizon returns, set three barriers:

- Upper barrier: price target (+5% for your use case)
- Lower barrier: stop loss (-3% or -5% depending on your risk tolerance)
- Time barrier: market close (for end-of-day MFE) or a maximum hold time (e.g., 120 minutes)

Label = whichever barrier is hit first. This is economically meaningful — the barriers ARE your trade parameters.

For your $\geq 5\%$ detection system specifically:

- LONG candidate: $\max(\text{high}[t : \text{close}]) \geq \text{entry}[t] * 1.05$
- SHORT candidate: $\min(\text{low}[t : \text{close}]) \leq \text{entry}[t] * 0.95$
- NO_TRADE: price stays within [-5%, +5%] until close

5.2 Class imbalance — expect severe

On any given day, approximately 15-40 of 438 NSE stocks will move $\geq 5\%$ intraday. Within those stocks, only the bars prior to the move are meaningful positive examples. Expect 2-5% positive class rate across the full dataset.

How to handle (in order of preference):

1. Precision-focused threshold tuning: optimize for precision $\geq 60\%$ at whatever recall the model achieves. Do NOT use SMOTE — it creates synthetic examples from interpolated microstructure states that don't exist in real markets.
2. Class weights in the loss function: set `class_weight='balanced'` in sklearn or `scale_pos_weight` in LightGBM. This is preferred over resampling.
3. Asymmetric cost matrix: a false positive (entering a trade that doesn't move) costs you transaction costs. A false negative (missing a 5% move) costs you opportunity. Decide which is worse for your strategy and set the cost ratio accordingly.
4. Calibration: after training, calibrate probabilities with Platt scaling or isotonic regression so $P(\text{LONG})=0.7$ actually corresponds to $\sim 70\%$ of examples being LONG. Required for meaningful position sizing.

6. Model Selection & Architecture

6.1 Recommended progression

Phase	Model	Rationale
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Now (1 day)	Logistic Regression (L1)	Establish cross-sectional baseline. Interpretable coefficients tell you which features are informative before getting more data.
Phase 2 (10-30 days)	Random Forest	Your current approach. Handles non-linearity without overfitting if depth is controlled. Provides feature importance.
Phase 3 (60+ days)	LightGBM	Native handling of missing values (important for early-session NaN features), faster training, better calibration than RF. Use SHAP for feature importance — more reliable than RF's impurity-based importance.
Phase 4 (6+ months)	Regime-aware ensemble	Separate models for opening volatility window (9:15-10:00), midday accumulation (10:00-13:00), and afternoon breakout (13:00-15:30). Combine with a meta-classifier.

6.2 Do NOT do these

- Do not use LSTM or transformer models until you have 2+ years of data. These models memorize regime — with < 6 months you will overfit to a single market condition.
- Do not use model complexity to compensate for insufficient data. A well-tuned LR on 60 days of good features will outperform a neural network on 1 day.
- Do not build live streaming infrastructure before your offline signal is validated on 6+ months of walk-forward tests. Every firm that built plumbing before proving signal wasted 6-12 months.
- Do not use SMOTE for class imbalance — it generates synthetic microstructure states that do not exist in real markets.

7. Validation Strategy

7.1 The honest single-day test (do this now)

Within-stock temporal split: for each stock, train on bars 1-250, test on bars 251-374. This tests whether the model predicts the afternoon from the morning — actual temporal generalization, not cross-sectional stock properties. If MCC on this test is near zero, you have confirmed the stock-characterization hypothesis.

7.2 Walk-forward validation (required once you have multi-day data)

The gold standard for intraday signal validation:

- Train on days 1-20, test on day 21
- Train on days 1-21, test on day 22
- Continue expanding window for each subsequent day
- Purge gap: exclude the last 60 bars of the training set to prevent look-ahead from 60-minute overlapping labels

- Embargo gap: exclude the first 30 bars of each test day to prevent information leakage at the boundary

7.3 Combinatorial Purged Cross-Validation (CPCV)

Lopez de Prado Chapter 12. For each fold boundary: (1) purge the N bars around the boundary where N = label horizon in bars (60 bars for 60-minute labels), (2) embargo a gap of 30 bars after each training set ends. This eliminates the look-ahead contamination from overlapping forward return labels that afflicts standard k-fold on financial time series.

7.4 Out-of-time validation (required before production)

Train on Jan-Sep, test on Oct-Dec. This is the only test that validates regime generalization. Without it you do not know if your signal survives a regime change (pre/post RBI policy, pre/post earnings season, bull vs bear, high-VIX vs low-VIX). Target: MCC in out-of-time test should be within 30% of in-sample MCC for the model to be production-worthy.

7.5 Deflated Sharpe Ratio

If you try 50 feature combinations before finding $MCC=0.31$, the true statistical significance of that result is much lower than it appears. Track every experiment you run. The Deflated Sharpe Ratio (DSR) accounts for multiple testing and gives you the true probability that your best result is not due to chance. Required discipline for any serious research process.

8. Metrics & Observability

8.1 Model evaluation metrics — complete stack

Metric	Priority	What It Tells You
MCC	KEEP	Correct for imbalanced multiclass. Your current primary metric.
Precision (LONG/SHORT separately)	ADD — HIGH	False positives cost money. Track precision by class at every decision threshold. Target: precision ≥ 0.60 at your operating point.
Recall (LONG/SHORT separately)	ADD — HIGH	Are you catching the actual movers? Track the precision-recall tradeoff curve, not just a single threshold.
Information Coefficient (IC)	ADD — HIGH	Spearman rank correlation between predicted probability and actual return. Sort all bars by P(LONG), bin into deciles, compute mean actual return per decile. If the top decile consistently outperforms the bottom decile, you have real signal. Standard metric for systematic equity PMs.

Calibration curve	ADD — HIGH	Does $P(\text{LONG})=0.70$ correspond to ~70% of examples actually being LONG? Uncalibrated probabilities make position sizing impossible.
Simulated P&L	ADD — HIGH	Enter at LONG signal, hold until 5% hit or close, apply 0.05% slippage each way. Calculate Sharpe, max drawdown, win rate, average hold time. A model with $\text{MCC}=0.15$ but negative Sharpe after slippage is worthless.
Feature importance stability	ADD — MEDIUM	Compute per fold. If top features change dramatically between folds, model is unstable and will degrade in production.
Feature importance over time	ADD — MEDIUM	Does <code>imbalance_top5</code> importance decay across walk-forward folds? Decay means the signal is regime-dependent.

8.2 Production observability (for when you go live)

Monitor	What to Watch For
Prediction distribution drift	If daily $\text{mean}(P(\text{LONG}))$ shifts from 0.35 to 0.55, your feature distribution has drifted — likely a market regime change. Trigger model retrain.
Feature distribution by stock	Flag stocks where feature values fall outside 3-sigma of training distribution. These are candidates for data quality issues or unusual market conditions.
Real-time IC	Compute rolling 10-day IC on your LONG signals. If IC drops below 0.03, signal is deteriorating. Below 0.01 for 3+ days = suspend trading, investigate.
Slippage vs model assumptions	Track actual entry slippage vs the 0.05% assumed in backtesting. If actual slippage is 0.15%, your simulated Sharpe is overstated.
Position concentration	Never have more than 20% of capital in a single sector. If the model learns a sector-specific pattern, concentration risk compounds.

9. Data Acquisition Path

9.1 Why more data is the single highest-leverage action

With 1 day of data: you can identify which stocks moved $\geq 5\%$ on May 29, 2026. With 60 days: you can test whether your microstructure signals persist across different market regimes. With 6 months: you can do meaningful walk-forward validation and regime analysis. With 2+ years: you can train regime-aware models and test across bull, bear, and range-bound conditions.

Target minimum: 60 trading days across at least one trending and one range-bound market period. $438 \text{ stocks} \times 60 \text{ days} = 26,280 \text{ stock-days}$.

9.2 Data sources for NSE L2 tick data

Source	Cost	Notes
Your current vendor	Already paying	First ask: do they have historical backfill for the same L2 DOM + tick format? You've already built the pipeline for their schema.
TickerPlant India	~INR 15-30K/mo	Institutional NSE L2 tick data. Covers all segments. Well-known vendor in Indian quant community.
True Data (TrueBeacon)	~INR 10-25K/mo	NSE tick and DOM data with good historical depth. Used by several Indian quant funds.
Refinitiv Tick History	USD pricing	Tick and L2 data for NSE. More expensive but institutional grade with long historical depth.
Zerodha Kite Connect	Low cost	1-min OHLCV only — no L2 DOM. Cannot rebuild your microstructure features. Useful for label validation and regime analysis only.

10. Direct Answers to Your 11 Questions

Q1: How to catch the contamination bug earlier?

Add a symbol identity test as a Stage 0 post-condition: sample 100 random rows from each output file, verify that every row's symbol column matches the filename. One assert, catches this class of bug in milliseconds. Also use property-based testing (Hypothesis library) on the splitter with randomised row-group orderings — the bug you hit is a hash-vs-sort order mismatch that only manifests with specific orderings.

Q2: Other assumption mismatches like the splitter bug?

Three areas to audit: (1) Timezone double-conversion — NSE publishes IST, your pipeline converts to UTC; verify no offset is applied twice with a roundtrip test. (2) Merge key precision — if Stage 3 DOM+tick join is timestamp-based, microsecond precision mismatch will silently drop rows. Verify row counts before and after merge. (3) Sort stability — anywhere you sort by timestamp, verify identical timestamps produce deterministic order using a secondary sort key (symbol or sequence number).

Q3: Cheapest way to validate trade inference accuracy?

Bar-level comparison is the right level — not trade-by-trade. For each symbol-minute: (a) inferred trade count vs real tick count, (b) inferred buy volume vs real buy volume, (c) inferred sell volume vs real sell volume. Report Pearson r and MAPE for each. Target: $r > 0.85$ on

counts, $r > 0.70$ on directional split. Budget: 2 hours. Full trade-by-trade matching is overkill given your features are all 1-min aggregates.

Q4: Testing if model learns stock-level properties vs temporal signals?

Three tests: (1) Compute per-stock mean(spread) and mean(volatility_5m). If these alone achieve $MCC > 0.15$ in a fresh LR, your $RF=0.31$ is a stock classifier. (2) Within-stock temporal split: train on first 70% of each stock's bars, test on last 30%. If MCC collapses to near zero, signal is cross-sectional not temporal. (3) Shuffle which stock each bar belongs to (keeping all features intact) — if MCC stays stable, features have no per-stock identity signal. All three take under 30 minutes to implement.

Q5: Better test for leakage vs benign cross-stock correlation?

Time-windowed shuffle: shuffle labels only within 30-minute windows (preserving intraday regime). If RF shuffled MCC drops to near zero under time-windowed shuffle but not global shuffle, the residual signal is from intraday macro regime, not stock-to-stock leakage. Also try alphabetical split: train on first 200 stocks, test on last 200, no temporal overlap. If MCC holds, you have cross-sectional signal that generalises to unseen stocks — a good result.

Q6: Is LR $MCC=0.1425$ genuine signal or artefact?

Cautiously real but not reliable on single-day data. LR's L1 regularization has correctly zeroed noise features. However, with 88k rows from a single day, all folds share the same intraday macro environment. True out-of-sample MCC on a genuinely held-out day from a different regime will likely be 0.06-0.10. The positive sign is encouraging — it is not noise. It is a weak cross-sectional signal that may or may not contain temporal predictive power. Multi-day walk-forward will resolve this.

Q7: Can order_cancel_rate be estimated from DOM snapshots?

Yes, approximately. DOM-diff approach: qty decrease at a level NOT followed by LTP change within ~1 tick time window = likely cancellation. $cancel_rate_proxy = \frac{\text{sum}(qty_drops_without_trade)}{\text{sum}(total_qty_changes)}$. Expected accuracy ~60-70% on NSE 20-level DOM. Confounds partial fills with cancellations at the same level. Implement as a DOM-diff feature rather than tick-derived feature. Better than all-zeros.

Q8: EOD MFE labels vs fixed-horizon improvement?

Abandon fixed-horizon, switch to MFE-to-close entirely. This is the right label for your stated goal. Run EOD MFE as your primary label for all future work. The fixed-horizon approach was a stepping stone — you've outgrown it now that you've articulated the actual trading objective.

Q9: Most efficient path to 3,000+ stock-days on NSE?

In order of speed: (1) Ask your current data vendor for historical backfill in the same format — you've already built the pipeline for their schema, zero additional engineering. (2) TickerPlant India or True Data for institutional NSE L2 historical data. (3) NSE's own data products for bhavcopy (limited to OHLCV). Target 60 trading days minimum before Phase 3 modelling. 438 stocks x 60 days = 26,280 stock-days. Prioritise days from different market regimes.

Q10: Priority order for resource-constrained team?

#	Priority
1	Validate trade inference against real ticks (2 hours) — gates everything else
2	Run stock-characterization tests (Q4) — determines if current RF signal is real
3	Acquire 60+ days of data across multiple market regimes
4	Fix vwap_distance, crossed book stocks, rebuild order_cancel_rate
5	Switch labels to MFE Triple Barrier
6	Add high-priority new features: micro-price, cumulative delta, book pressure, time-of-day, opening range
7	Implement walk-forward validation replacing stock-level CV
8	Add IC, calibration, P&L simulation to evaluation stack
9	Upgrade to LightGBM with SHAP feature importance
10	Live streaming infrastructure (last — do not build until offline signal proven on 6+ months walk-forward)

Q11: Is $MCC=0.31$ from RF 'too good to be true'?

Yes, be suspicious. Realistic MCC bounds for a generalisable microstructure signal on NSE equities: LR baseline 0.04-0.10, tree ensemble 0.08-0.18, tuned LightGBM 0.12-0.22, regime-aware ensemble 0.15-0.28. $MCC=0.31$ sits above the realistic upper bound. Given spread and volatility_5m as top RF features — both near-static stock properties — the most likely explanation is a stock-type classifier achieving good cross-sectional MCC but near-zero temporal MCC. Run the within-stock temporal CV test. It will answer this definitively.

11. Essential Reading

These are non-negotiable references for the approach you are building:

- Lopez de Prado, M. (2018). Advances in Financial Machine Learning. Wiley. — Chapter 3 (Triple Barrier Labels), Chapter 7 (Feature Importance), Chapter 12 (Combinatorial Purged CV). Your instincts are already close to this framework.

- Cartea, A., Jaimungal, S., Penalva, J. (2015). Algorithmic and High-Frequency Trading. Cambridge. — Chapters 6-8 on order flow and microstructure signals.
- Avellaneda, M. & Stoikov, S. (2008). High-frequency trading in a limit order book. Quantitative Finance. — The foundational paper for microstructure-based signal construction.
- Easley, D., Lopez de Prado, M., O'Hara, M. (2012). Flow Toxicity and Liquidity in a High-Frequency World. Review of Financial Studies. — The original VPIN paper.
- Kissell, R. (2013). The Science of Algorithmic Trading and Portfolio Management. Academic Press. — Chapters on transaction cost analysis and slippage modelling for realistic P&L simulation.

Final Word

Your engineering discipline is well above average for this stage of a project. The contamination handling, the formal review loops, the documentation — these habits separate serious quant teams from hobbyists. Do not lose them as you scale.

The path forward is clear: validate your trade inference, confirm or refute the stock-characterization hypothesis, switch to MFE labels, acquire 60+ days of data, and let the model tell you which features matter. The microstructure signals you are building around — cumulative delta, iceberg detection, book pressure, order arrival rate — are genuinely the right signals for intraday move detection. You are looking in the right place. Now build the rigor around it.